



Java TreeMap

[< Previous](#)[Next >](#)

Java TreeMap

A **TreeMap** is a collection that stores key/value pairs **in sorted order by key**.

It is part of the `java.util` package and implements the `Map` interface.

Tip: Unlike `HashMap`, which does not maintain order, **TreeMap** keeps its keys sorted.

Create a TreeMap

Create a **TreeMap** that stores **String** keys and **String** values:

Example

[Get your own Java Server](#)

```
import java.util.TreeMap; // Import the TreeMap class

TreeMap<String, String> capitalCities = new TreeMap<>();
```



Add Items

Use the `put()` method to add key/value pairs:

Example

```
import java.util.TreeMap;

public class Main {
    public static void main(String[] args) {
        TreeMap<String, String> capitalCities = new TreeMap<>();
        capitalCities.put("England", "London");
        capitalCities.put("India", "New Dehli");
        capitalCities.put("Austria", "Wien");
        capitalCities.put("Norway", "Oslo");
        capitalCities.put("Norway", "Oslo"); // Duplicate
        capitalCities.put("USA", "Washington DC");

        System.out.println(capitalCities);
    }
}
```

Try it Yourself »

Output: The keys are sorted alphabetically (e.g., {Austria=Wien, England=London, India=New Dehli, Norway=Oslo, USA=Washington DC}).

Note: Duplicates like "Norway" will only appear once.



Example

```
capitalCities.get("England");
```

[Try it Yourself »](#)

Remove Items

Use `remove()` to delete a key/value pair by key:

Example

```
capitalCities.remove("England");
```

[Try it Yourself »](#)

Use `clear()` to remove all items:

Example

```
capitalCities.clear();
```

[Try it Yourself »](#)



Use `size()` to count the number of key/value pairs:

Example

```
capitalCities.size();
```

Try it Yourself »

Note: The size only counts unique keys. If a key is added more than once, only the latest value is kept.

Loop Through a TreeMap

Loop through the items of a `TreeMap` with a for-each loop.

Note: Use the `keySet()` method if you only want the keys, and use the `values()` method if you only want the values:

Example

```
// Print keys
for (String i : capitalCities.keySet()) {
    System.out.println(i);
}
```

Try it Yourself »



```
// Print values
for (String i : capitalCities.values()) {
    System.out.println(i);
}
```

[Try it Yourself »](#)

Example

```
// Print keys and values
for (String i : capitalCities.keySet()) {
    System.out.println("key: " + i + " value: " + capitalCities.get(i));
}
```

[Try it Yourself »](#)

TreeMap vs HashMap

Feature	HashMap	TreeMap
Order	No guaranteed order	Sorted by keys
Null Keys	Allows one null key	Does not allow null keys
Performance	Faster (no sorting)	Slower (maintains sorted order)

Tip: Use [HashMap](#) for performance, and [TreeMap](#) when you need sorted keys.



From Java 10, you can use the **var** keyword to declare a **TreeMap** variable without writing the type twice. The compiler figures out the type from the value you assign.

This makes code shorter, **but many developers still use the full type for clarity**. Since **var** is valid Java, you may see it in other code, so it's good to know that it exists:

Example

```
// Without var
TreeMap<String, String> capitalCities = new TreeMap<String, String>();

// With var
var capitalCities = new TreeMap<String, String>();
```

[Try it Yourself »](#)

The Map Interface

Note: Sometimes you will see both **Map** and **TreeMap** in Java code, like this:

```
import java.util.Map;
import java.util.TreeMap;

Map<String, String> capitalCities = new TreeMap<>();
```

[Try it Yourself »](#)

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

It works the same way, but some developers prefer this style because it gives them more flexibility to change the type later.

[◀ Previous](#)[Sign in to track progress](#)[Next ▶](#)

ADVERTISEMENT



ADVERTISEMENT

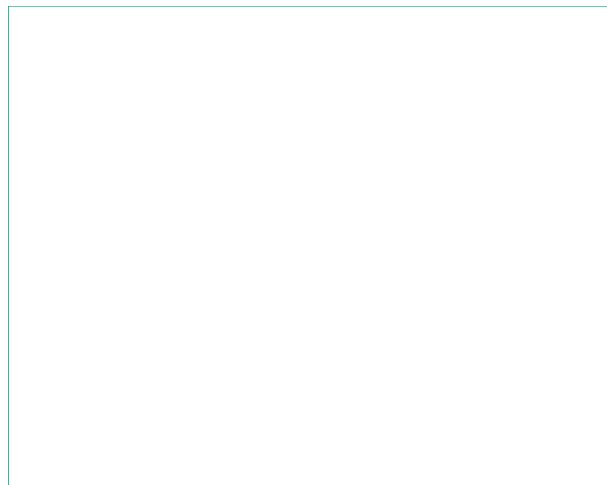


REMOVE ADS

ADVERTISEMENT

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#) [JAVASCRIPT](#) [SQL](#) [PYTHON](#) [JAVA](#) [PHP](#) [HOW TO](#) [W3.CSS](#) [C](#)

ADVERTISEMENT



PLUS

SPACES

GET CERTIFIED

FOR TEACHERS

FOR BUSINESS

CONTACT US

Top Tutorials

W3.CSS Tutorial

Bootstrap Tutorial

PHP Tutorial

Java Tutorial

C++ Tutorial

jQuery Tutorial

Top References

HTML Reference

CSS Reference

JavaScript Reference

SQL Reference

Python Reference

W3.CSS Reference

Bootstrap Reference

PHP Reference

HTML Colors

Java Reference

AngularJS Reference

jQuery Reference

Top Examples

HTML Examples

CSS Examples

JavaScript Examples

How To Examples

SQL Examples

Python Examples

W3.CSS Examples

Bootstrap Examples

PHP Examples

Java Examples

XML Examples

jQuery Examples

Get Certified

HTML Certificate

CSS Certificate

JavaScript Certificate

Front End Certificate

SQL Certificate

Python Certificate

PHP Certificate

jQuery Certificate

Java Certificate

C++ Certificate

C# Certificate

XML Certificate



FORUM

ABOUT

ACADEMY

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning.

Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our [terms of use](#), [cookie and privacy policy](#).

Copyright 1999-2025 by Refsnes Data. All Rights Reserved. [W3Schools](#) is Powered by [W3.CSS](#).